

Robotics With a Twist

$$\text{DOF}_r = 6(n_{\text{links}} - 1 - n_{\text{joints}}) + \sum_i^{n_{\text{joints}}} \text{DOF}_i$$

$$\text{Homo.}\{\text{Point: } [\mathbf{p}^T \ 1]^T \ \text{Vector: } [\mathbf{v}^T \ 0]^T\}$$

$${}^0\mathbf{T}_n(\boldsymbol{\theta}) = \exp(\boldsymbol{\Xi}_1\theta_1) \cdots \exp(\boldsymbol{\Xi}_n\theta_n) {}^0\mathbf{T}_n(\mathbf{0})$$

$$x(t) = \sum_{i=0}^n a_i t^i \quad t_0 \leq t \leq t_1$$

$${}^0\mathbf{R}_1 {}^1\mathbf{R}_2 = {}^0\mathbf{R}_2 \implies \mathbf{R}_t(t) = {}^0\mathbf{R}_1 \mathbf{R}(\mathbf{k}\theta(t))$$

Angle Axis

$$\mathbf{R} = \exp(\mathbf{S}(\mathbf{k}\theta))$$

$$= \mathbf{I} + s\theta \mathbf{S}(\mathbf{k}) + \mathbf{S}^2(\mathbf{k})(1 - c\theta)$$

$$= \mathbf{I}c\theta + s\theta \mathbf{S}(\mathbf{k}) + \mathbf{k}\mathbf{k}^T(1 - c\theta)$$

$$c\theta = \frac{\text{Tr}(\mathbf{R}) - 1}{2} \quad \mathbf{k} = \frac{1}{2s\theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

$$s\theta = \pm \frac{1}{2} \left(\begin{array}{c} (r_{32} - r_{23})^2 \\ + (r_{13} - r_{31})^2 \\ + (r_{21} - r_{12})^2 \end{array} \right)^{1/2}$$

Homogeneous Transform $\longrightarrow$ Twist

Let $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{d} \\ \mathbf{0}^T & 1 \end{bmatrix}$ known.

if $\mathbf{R} = \mathbf{I}$ and $\mathbf{d} \neq \mathbf{0}$:

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} \quad \boldsymbol{\Xi} = \begin{bmatrix} \mathbf{0} & \mathbf{d} \\ \mathbf{0}^T & 0 \end{bmatrix}$$

elif $\mathbf{R} \neq \mathbf{I}$ and $\mathbf{d} = \mathbf{0}$:

Extract $\mathbf{k}$ and θ using angle-axis

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{0} \\ \mathbf{k} \end{bmatrix} \theta \quad \boldsymbol{\Xi} = \begin{bmatrix} \mathbf{S}(\mathbf{k}\theta) & \mathbf{0} \\ \mathbf{0}^T & 0 \end{bmatrix}$$

else:

Extract $\mathbf{k}$ and θ using angle-axis

$$\mathbf{v} = \left[\frac{1}{\theta} (\mathbf{I} - \mathbf{R}) \mathbf{S}(\hat{\mathbf{k}}) + \hat{\mathbf{k}}\hat{\mathbf{k}}^T \right]^{-1} \mathbf{d}$$

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{v} \\ \mathbf{k} \end{bmatrix} \theta \quad \boldsymbol{\Xi} = \begin{bmatrix} \mathbf{S}(\mathbf{k}\theta) & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix}$$

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Homogeneous Transform $\longrightarrow$ Screw

Let $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{d} \\ \mathbf{0}^T & 1 \end{bmatrix}$ known.

Do HT $\longrightarrow$ Twist, and get $\mathbf{v}, \mathbf{k}, \theta$. Then

$$h = (\mathbf{k}^T \mathbf{v}) / \theta \quad \mathbf{q} = (\mathbf{k} \times \mathbf{v}) / \theta$$

Screw $\longrightarrow$ Twist

Let $\mathbf{q}, \mathbf{k}, h$, and θ known.

if $h = \infty$:

$$\mathbf{v} = \mathbf{q} \times \mathbf{k} \quad \boldsymbol{\xi} = \begin{bmatrix} \mathbf{v} \\ \mathbf{0} \end{bmatrix} \quad \boldsymbol{\Xi} = \begin{bmatrix} \mathbf{0} & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix}$$

else:

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{q} \times \mathbf{k} + \mathbf{k}h \\ \mathbf{k} \end{bmatrix} \theta = \begin{bmatrix} \mathbf{v} \\ \mathbf{k} \end{bmatrix} \theta \quad \boldsymbol{\Xi} = \boldsymbol{\xi}^\wedge$$

Twist $\longrightarrow$ Screw

Let $\boldsymbol{\Xi} = \begin{bmatrix} \mathbf{S}(\mathbf{k}\theta) & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix}$ known.

Extract $\mathbf{k}, \theta$, and $\mathbf{v}$ from $\boldsymbol{\Xi}$

$$h = (\mathbf{k}^T \mathbf{v}) / \theta \quad \mathbf{q} = (\mathbf{k} \times \mathbf{v}) / \theta$$

Screw $\longrightarrow$ Homogeneous Transform

Let $\mathbf{q}, \mathbf{k}, h$, and θ known.

if $h = \infty$:

$$\mathbf{T} = \begin{bmatrix} \mathbf{I} & \mathbf{v} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

else:

$$\mathbf{T} = \begin{bmatrix} \exp(\mathbf{S}(\mathbf{k}\theta)) & (\mathbf{I} - \exp(\mathbf{S}(\mathbf{k}\theta))\mathbf{q} + h\theta\mathbf{k}) \\ \mathbf{0}^T & 1 \end{bmatrix}$$

Twist $\longrightarrow$ Homogeneous Transform

Let $\boldsymbol{\Xi} = \begin{bmatrix} \mathbf{S}(\mathbf{k}\theta) & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix}$ known.

$$\mathbf{T} = \exp(\boldsymbol{\Xi})$$

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Velocity and updates

$${}^s\dot{\mathbf{R}}_b = \mathbf{S}({}^s\boldsymbol{\omega}_{sb}) {}^s\mathbf{R}_b = {}^s\mathbf{R}_b \mathbf{S}({}^b\boldsymbol{\omega}_{sb})$$

$$\begin{aligned} {}^s\mathbf{R}_{sb}(t+dt) &= \exp(\mathbf{S}({}^s\boldsymbol{\omega}_{sb}dt)) {}^s\mathbf{R}_b(t) \\ &= {}^s\mathbf{R}_b(t) \exp(\mathbf{S}({}^b\boldsymbol{\omega}_{sb}dt)) \end{aligned}$$

$${}^s\dot{\mathbf{T}}_b = {}^s\boldsymbol{\Xi}_{ss} {}^s\mathbf{T}_b = {}^s\mathbf{T}_b {}^b\boldsymbol{\Xi}_{sb}$$

$$\begin{aligned} {}^s\mathbf{T}_b(t+dt) &= \exp({}^s\boldsymbol{\Xi}_{ss}dt) {}^s\mathbf{T}_b(t) \\ &= {}^s\mathbf{T}_b(t) \exp({}^b\boldsymbol{\Xi}_{sb}dt) \end{aligned}$$

$$\begin{aligned} \dot{\mathbf{q}} &= \frac{1}{2} \begin{bmatrix} -\mathbf{q}^T \\ -\mathbf{S}(\mathbf{q}) + \mathbf{q}_0 \mathbf{I} \end{bmatrix} \boldsymbol{\omega} = \frac{1}{2} \begin{bmatrix} 0 & -\boldsymbol{\omega}^T \\ \boldsymbol{\omega} & \mathbf{S}(\boldsymbol{\omega}) \end{bmatrix} \mathbf{q} \\ \mathbf{q}(t+dt) &= \mathbf{q}(t) + \dot{\mathbf{q}}(t)dt \end{aligned}$$

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Spatial Jacobian $\mathbf{J}_s$

$$\begin{bmatrix} {}^0\mathbf{d}_{00} & {}^0\boldsymbol{\omega}_{0n}^T \end{bmatrix}^T = \mathbf{J}_s \dot{\boldsymbol{\theta}}$$

$$\mathbf{J}_s = [\boldsymbol{\xi}_1 \quad \boldsymbol{\xi}'_2 \quad \cdots \quad \boldsymbol{\xi}'_n]$$

$$\boldsymbol{\xi}'_{is} = A[\exp(\boldsymbol{\Xi}_1\theta_1) \cdots \exp(\boldsymbol{\Xi}_{i-1}\theta_{i-1})]\boldsymbol{\xi}_i$$

Velocity Jacobian $\mathbf{J}_v$

$$\begin{bmatrix} {}^0\dot{\mathbf{d}}_{0n}^T & {}^0\boldsymbol{\omega}_{0n}^T \end{bmatrix}^T = \mathbf{J}_v \dot{\boldsymbol{\theta}} \quad \text{prism.} \begin{bmatrix} {}^0\mathbf{z}_{i-1} \\ \mathbf{0} \end{bmatrix}$$

$$\mathbf{J}_v = \begin{bmatrix} {}^0\mathbf{z}_0 \times {}^0\mathbf{d}_{0n} & \cdots & {}^0\mathbf{z}_{n-1} \times {}^0\mathbf{d}_{n-1,n} \\ {}^0\mathbf{z}_0 & \cdots & {}^0\mathbf{z}_{n-1} \end{bmatrix}$$

Body Jacobian $\mathbf{J}_b$

$$\begin{bmatrix} {}^n\mathbf{d}_{0n} & {}^n\boldsymbol{\omega}_{0n}^T \end{bmatrix}^T = \mathbf{J}_b \dot{\boldsymbol{\theta}}$$

$$\mathbf{J}_b = [\boldsymbol{\xi}'_1 \quad \cdots \quad \boldsymbol{\xi}'_n]$$

$$\boldsymbol{\xi}'_{ib} = A^{-1}[\exp(\boldsymbol{\Xi}_i\theta_i) \cdots \exp(\boldsymbol{\Xi}_n\theta_n) {}^0\mathbf{T}_n(\mathbf{0})]\boldsymbol{\xi}_i$$

Jacobian Conversions

$${}^{\alpha}\boldsymbol{\xi}_{\beta} = \begin{bmatrix} {}^{\alpha}\boldsymbol{v}_{\beta} \\ {}^{\alpha}\boldsymbol{\omega}_{\beta} \end{bmatrix} \quad (\alpha, \beta) \in \left\{ \begin{array}{ll} (s, sb) & \text{trad.} \\ (b, sb) & \text{body} \\ (s, ss) & \text{spatial} \end{array} \right\}$$

$$\boldsymbol{J}_v = \begin{bmatrix} {}^s\boldsymbol{R}_b & \mathbf{0} \\ \mathbf{0} & {}^s\boldsymbol{R}_b \end{bmatrix} \boldsymbol{J}_b \quad \boldsymbol{J}_s = A({}^s\boldsymbol{T}_n) \boldsymbol{J}_b$$

$${}^s\boldsymbol{\xi}_{ss} = \begin{bmatrix} {}^s\boldsymbol{R}_b & \boldsymbol{S}({}^s\boldsymbol{d}_{sb}) {}^s\boldsymbol{R}_b \\ \mathbf{0} & \boldsymbol{R}_b \end{bmatrix} {}^b\boldsymbol{\xi}_{sb}$$

$$= A({}^s\boldsymbol{T}_b) {}^b\boldsymbol{\xi}_{sb}$$

$$A({}^s\boldsymbol{T}_b) = \begin{bmatrix} {}^s\boldsymbol{R}_b & \boldsymbol{S}({}^s\boldsymbol{d}_{sb}) {}^s\boldsymbol{R}_b \\ \mathbf{0} & {}^s\boldsymbol{R}_b \end{bmatrix}$$

$$A^{-1}({}^s\boldsymbol{T}_b) = \begin{bmatrix} {}^s\boldsymbol{R}_b^T & -{}^s\boldsymbol{R}_b^T \boldsymbol{S}({}^s\boldsymbol{d}_{sb}) \\ \mathbf{0} & {}^s\boldsymbol{R}_b^T \end{bmatrix}$$

$$= A({}^s\boldsymbol{T}_b^{-1})$$

Singular Value Decomposition

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T \quad \boldsymbol{A}^{\dagger} = \boldsymbol{V}\boldsymbol{\Sigma}^{\dagger}\boldsymbol{U}^T = (\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T$$

$$\kappa(\boldsymbol{A}) = \frac{\sigma_{\max}(\boldsymbol{A})}{\sigma_{\min}(\boldsymbol{A})} \in [1, \infty) \quad \kappa^{-1} \in [0, 1]$$

$$\boldsymbol{A}\boldsymbol{N} = \boldsymbol{A}[\boldsymbol{I} - \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{A}^T)^{\dagger}\boldsymbol{A}]$$

$$\dot{\boldsymbol{\theta}} = \boldsymbol{W}_{\text{in}}(\boldsymbol{W}_{\text{out}}\boldsymbol{J}\boldsymbol{W}_{\text{in}})^{\dagger}\boldsymbol{W}_{\text{out}}\boldsymbol{\xi}_{\text{des}}$$

Properties of Vector Quantities

$$\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) = (\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{c}$$

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = (\boldsymbol{a} \cdot \boldsymbol{c})\boldsymbol{b} - (\boldsymbol{a} \cdot \boldsymbol{b})\boldsymbol{c}$$

$$\boldsymbol{S}^2(\boldsymbol{k}) = \boldsymbol{k}\boldsymbol{k}^T - \boldsymbol{I} \quad \text{if} \quad \|\boldsymbol{k}\| = 1$$

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Conservation of Power (Statics)

$$\begin{aligned} p &= \dot{\boldsymbol{\theta}}^T \boldsymbol{\tau} \\ &= -\dot{\boldsymbol{\theta}}^T \mathbf{J}_v^T {}^0 \mathbf{w}_n \\ &= -\dot{\boldsymbol{\theta}}^T \mathbf{J}_b^T {}^n \mathbf{w}_n \end{aligned}$$

Gravity Balance

$$\begin{aligned} \boldsymbol{\tau} &= -\sum_{i=1}^n \mathbf{J}_{vi}^T \left[\mathbf{S}({}^0 \mathbf{r}_{i,COMi}) \right] {}^0 \mathbf{g} m_i \\ &= -\sum_{i=1}^n \mathbf{J}_{bi}^T \left[\mathbf{S}({}^i \mathbf{r}_{i,COMi}) \right] {}^i \mathbf{g} m_i \end{aligned}$$

Dynamics

$$\begin{aligned} \mathbf{L}_{0i} &= \mathbf{q}_{0,i} \times m_i (\boldsymbol{\omega}_{0i} \times \mathbf{q}_{0,i}) = \mathbf{I}_{0i} \boldsymbol{\omega}_{0i} \\ &= \left[\sum_{k=1}^N (m_{ik} \|\mathbf{q}_{0,ik}\|^2 \mathbf{I} - m_{ik} \mathbf{q}_{0,ik} \mathbf{q}_{0,ik}^T) \right] \boldsymbol{\omega}_{0i} \\ \mathbf{I}_{0i} &= \mathbf{I}_i + m_i (\|\mathbf{r}_{01}\|^2 \mathbf{I} - \mathbf{r}_{0i} \mathbf{r}_{0i}^T) \\ {}^0 \mathbf{I}_i &= {}^0 \mathbf{R}_i {}^i \mathbf{I}_i {}^0 \mathbf{R}_i^T \\ {}^0 \mathbf{w}_i &= \begin{bmatrix} m_i \mathbf{I} & \mathbf{0} \\ \mathbf{0} & {}^0 \mathbf{I}_i \end{bmatrix} {}^0 \dot{\boldsymbol{\xi}}_{0i} + \begin{bmatrix} \mathbf{0} \\ {}^0 \boldsymbol{\omega}_{0i} \times {}^0 \mathbf{I}_i {}^0 \boldsymbol{\omega}_{0i} \end{bmatrix} \\ {}^i \mathbf{w}_i &= \begin{bmatrix} m_i \mathbf{I} & \mathbf{0} \\ \mathbf{0} & {}^i \mathbf{I}_i \end{bmatrix} {}^i \dot{\boldsymbol{\xi}}_{0i} + \begin{bmatrix} {}^i \boldsymbol{\omega}_{0i} \times m_i {}^i \mathbf{v}_{0i} \\ {}^i \boldsymbol{\omega}_{0i} \times {}^i \mathbf{I}_i {}^i \boldsymbol{\omega}_{0i} \end{bmatrix} \\ \boldsymbol{\xi}_{0i}(t + \delta t) &\approx \boldsymbol{\xi}_{0i}(t) + \dot{\boldsymbol{\xi}}_{0i}(t) \delta t \end{aligned}$$

Lagrangian Dynamics

$$\mathcal{L} = \mathcal{T} - \mathcal{V} \quad \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} - \frac{\partial \mathcal{L}}{\partial \theta_i} = \Upsilon_i = \tau_i - \beta_i \dot{\theta}_i$$

$$\mathcal{T}_i = \frac{1}{2} \dot{\theta}^T \mathbf{J}_{b,Q_i}^T \mathbb{M}_i \mathbf{J}_{b,Q_i} \dot{\theta} \quad K_r = \frac{1}{2} \omega_{0i} \cdot \mathbf{I}_{0i} \omega_{0i}$$

$$\mathcal{T} = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \mathbf{M}_{ij} \dot{\theta}_i \dot{\theta}_j \quad \mathbb{M}_i = \begin{bmatrix} m_i \mathbf{I} & \mathbf{0} \\ \mathbf{0} & i \mathbf{I}_i \end{bmatrix}$$

$$\mathbf{M}(\theta) = \sum_{i=1}^n \mathbf{J}_{b,Q_i}^T \mathbb{M}_i \mathbf{J}_{b,Q_i} = \{\mathbf{M}_{ij}\}$$

$$\mathcal{V}_i = m_i \| \mathbf{g} \| h(\theta) \quad \mathcal{V} = \sum_{i=1}^n \mathcal{V}_i$$

$$\mathbf{N}_i = \partial \mathcal{V} / \partial \theta_i + \beta_i \dot{\theta}_i \quad \mathbf{C}_{ij} = \sum_{k=1}^n \Gamma_{ijk} \dot{\theta}_k$$

$$\Gamma_{ijk} = \frac{1}{2} \left(\frac{\partial \mathbf{M}_{ij}}{\partial \theta_k} + \frac{\partial \mathbf{M}_{ik}}{\partial \theta_j} - \frac{\partial \mathbf{M}_{kj}}{\partial \theta_i} \right)$$

$$\mathbf{M}(\theta) \ddot{\theta} + \mathbf{C}(\theta, \dot{\theta}) \dot{\theta} + \mathbf{N}(\theta, \dot{\theta}) = \boldsymbol{\tau}$$

$$\mathbf{M}(\theta) \ddot{\theta} + (\mathbf{C}(\theta, \dot{\theta}) + \mathbf{B}) \dot{\theta} = \boldsymbol{\tau}' = \boldsymbol{\tau} - \mathbf{G}(\theta)$$

Parallel Robots

$$\begin{bmatrix} \mathbf{J}_1 & -\mathbf{J}_2 & & \\ & & \ddots & \\ & & & \mathbf{J}_{m-1} & -\mathbf{J}_m \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \vdots \\ \dot{\theta}_m \end{bmatrix} = \mathbf{0}$$

$$\begin{bmatrix} \dot{\theta}_a^T & \dot{\theta}_p^T \end{bmatrix}^T = \mathbf{B}^{-1} \dot{\theta} \quad \mathbf{C}_a \dot{\theta}_a + \mathbf{C}_p \dot{\theta}_p = \mathbf{0}$$

if $\mathbf{C}_p \mathbf{C}_p^\dagger \mathbf{C}_a = \mathbf{C}_a$:

Then build $\boldsymbol{\xi}$ from $\dot{\theta}_a$ from an active joint, and the rest using

$$\dot{\theta}_p = -\mathbf{C}_p^\dagger \mathbf{C}_a \dot{\theta}_a = \mathbf{A} \dot{\theta}_a$$

else:

panic.

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